

Question	Marking Guidance	Mark	Comments																																
03.1	1 mark for each correct column;;;; <table> <tr> <th></th><th colspan="3">Biological molecule</th></tr> <tr> <th>Feature</th><th>Glycogen</th><th>Sucrose</th><th>DNA</th></tr> <tr> <td>Contains glucose</td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>Contains galactose</td><td></td><td></td><td></td></tr> <tr> <td>Contains fructose</td><td></td><td>✓</td><td></td></tr> <tr> <td>Contains phosphate</td><td></td><td></td><td>✓</td></tr> <tr> <td>Formed by condensation reaction(s)</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Is a polymer</td><td>✓</td><td></td><td>✓</td></tr> </table>		Biological molecule			Feature	Glycogen	Sucrose	DNA	Contains glucose	✓	✓		Contains galactose				Contains fructose		✓		Contains phosphate			✓	Formed by condensation reaction(s)	✓	✓	✓	Is a polymer	✓		✓	3 (3 x AO1)	
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03.2	Correct answer of 14/14.2 = 1 mark ;	1 (1 x AO2)	Accept numbers that round to 14.2																																

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03.3	1. (X) Galactose/inhibitor binds to active site; 2. (X so) slows/reduces/inhibits lactose hydrolysis; 3. (Y) all/more lactose/substrate is hydrolysed OR (Y) (more) lactose hydrolysed so more glucose/>100 (produced); 4. (Z) no lactose/substrate (so no product) OR (Z) Galactose is not hydrolysed by lactase OR (Z) Galactose is not a (lactase) substrate;	4 (4 x AO2)	2, 3 and 4 Accept digest/breakdown for hydrolysis 2. Reject 'stops' hydrolysis 3. Accept faster substrate hydrolysis for 'all/more' 4. Accept (Z) (only) lactose is the substrate

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03.4	1. Increase lactose/substrate (concentration); 2. More product/glucose (is formed) OR Increased rate of reaction;	2 (2 x AO3)	
